SAHARA: Safety-Asymmetric, Privacy-Preserving On-Device Sound Awareness for Deaf and Hard-of-Hearing Users in Indian Homes

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Abstract. Deaf and hard-of-hearing (DHH) people miss safety-critical sounds such as smoke alarms and sirens, yet sound-recognition aids treat all classes alike, often depend on the cloud, and are rarely evaluated outside Western homes. We present SAHARA, a fully on-device system that runs a frozen YAMNet encoder and a 296k-parameter classifier in an Android foreground service and drives a Rs. 870 ESP32 haptic wristband. SAHARA encodes safety asymmetry at every stage: a cost-weighted loss, precision-floored threshold calibration, an asymmetric temporal decision rule with closed-form false-alarm and latency bounds, and class-coded vibration. No audio ever leaves device memory. We also contribute SAHARA-IN, 123 clips recorded in Indian homes and streets with a recorder-disjoint protocol. On 698 held-out public clips the baseline reaches 0.651 mAP, and per-class analysis shows that the classes with the least data, smoke alarm and baby cry, are the weakest. **[TBD: one sentence with the final test mAP, SAHARA-IN result and false alarms per hour]**

Keywords: assistive technology, deaf and hard of hearing, sound event detection, on-device machine learning, haptic feedback, cost-sensitive learning

1 Introduction

More than 430 million people worldwide live with disabling hearing loss [23], and the 2011 Census of India recorded over five million people with a hearing disability [19]. For them, everyday acoustic events carry information that hearing people take for granted: a doorbell signals a visitor, a pressure-cooker whistle or microwave beep signals that food needs attention, and a smoke alarm or approaching siren signals danger. Surveys of DHH users consistently rank alarms and safety sounds as the most important to be notified about, and they prefer glanceable, wearable and private notifications [1, 3].

Deep audio embeddings trained on AudioSet [5] have made sound recognition on phones and watches practical [11–13, 16]. Three gaps remain for real-world assistive deployment. First, existing systems optimise a symmetric objective (accuracy or mean average precision), although the costs are deeply asymmetric: a missed fire alarm can be fatal, while a false doorbell alert is only an annoyance. Second, many prototypes stream audio or features to a server, which is unacceptable for an always-listening device in the home. Third, public benchmarks come mostly from Western acoustic environments. Indian homes add pressure-cooker whistles, temple bells, street hawkers, ceiling fans and monsoon rain, and their doorbells are often melodic chimes rather than two-tone buzzers.

This paper presents SAHARA (Sound-based Alert and Haptic Assistance for Real-world Awareness), an end-to-end system that addresses these gaps. Our contributions are:

1. **A safety-asymmetric detection stack.** We encode unequal error costs at four points: training (positive-weighted binary cross-entropy), calibration (per-class thresholds that maximise recall subject to a precision floor for safety-critical classes), streaming decision (a 1-of-3 rule for safety-critical classes and a 2-of-3 rule with cooldown for the rest), and actuation (distinct haptic codes, with the "temporal-three" fire-alarm rhythm [9] reserved for fire). We derive closed-form false-alarm and latency bounds for the decision rule (Sect. 4.3).
2. **Privacy by construction.** All inference runs on the phone. Raw audio stays in a ring buffer in RAM, the release build requests no Internet permission, and the phone-to-wristband protocol carries only a class identifier and an intensity byte (Sect. 3).
3. **SAHARA-IN**, a field-recorded set of Indian domestic and street audio with a recorder-disjoint evaluation protocol, alongside a pipeline that overlays target events on Indian noise from iNoise [15] at -5 to 10 dB SNR (Sect. 6).
4. **An open, low-cost wristband**: an ESP32 BLE peripheral with a coin vibration motor, a bill of materials of about Rs. 870 and battery-aware firmware (Sect. 5).

2 Related Work

Sound Awareness for DHH Users. Early work studied how to visualise non-speech sound [17] and how DHH users want to be notified [1, 3]. Jain et al. deployed HomeSound, an in-home sound awareness system, and found that false alerts quickly erode trust [11]; SoundWatch moved recognition to a smartwatch and analysed the trade-offs between accuracy, latency and battery life [12]. ProtoSound personalises recognition from a few user-recorded examples with prototypical networks [13, 21], and Goodman et al. studied user-driven personalisation [6]. These systems treat all classes symmetrically when they train and set thresholds, and they were evaluated in North American homes. SAHARA is complementary: it makes the cost asymmetry explicit throughout the pipeline and targets Indian acoustic conditions.

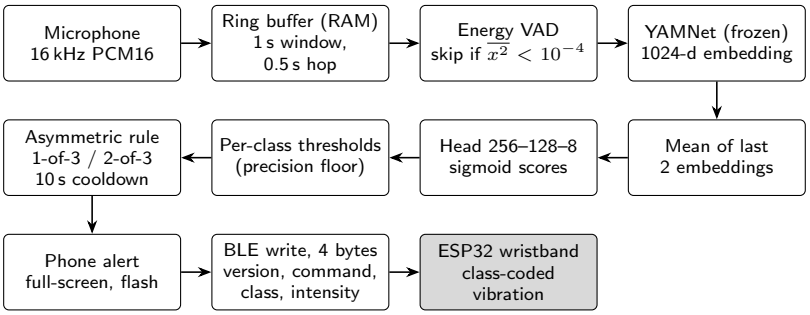


Fig. 1. SAHARA pipeline. The *white* blocks run on the phone inside an Android foreground service, and the *shaded* block is the wristband; only a class identifier and an intensity byte cross the Bluetooth link

Audio Embeddings and On-Device Inference. VGGish [7], PANNs [14] and YAMNet [20], a MobileNetV1 [8] trained on AudioSet, provide transferable embeddings. Ubicoustics showed that such models transfer to real-world activity recognition with little data [16]. We use YAMNet because it is small enough for continuous inference on mid-range Android phones and has a mature TensorFlow Lite path with integer quantisation [10].

Datasets and Cost-Sensitive Detection. FSD50K [4], AudioSet [5] and DESED [22] are standard sources of domestic sound events; Mesaros et al. give a tutorial on sound event detection and its metrics [18]. Cost-sensitive learning is well established [2], but to our knowledge it has not been combined with constrained threshold calibration and asymmetric temporal decision rules in an assistive sound-awareness system.

3 System Overview

Figure 1 shows the pipeline. A native Kotlin foreground service captures 16 kHz mono audio, so detection continues with the screen off. Samples enter a fixed 1 s ring buffer that advances every 0.5 s. A mean-square energy gate (10^{-4}) skips silent windows to save power. Each remaining window (the 0.975 s YAMNet patch) is embedded by a frozen YAMNet into a 1024-dimensional vector. The mean of the two most recent embeddings feeds a classifier head (Dense 256, ReLU, dropout 0.3; Dense 128, ReLU, dropout 0.3; Dense 8, sigmoid) that outputs an independent score for each of eight classes: smoke alarm, doorbell, siren, knocking, dog bark, baby cry, glass break and appliance beep. The head has $1024 \times 256 + 256 + 256 \times 128 + 128 + 128 \times 8 + 8 = 296{,}328$ parameters, about 1.2 MB in 32-bit floats. A Flutter interface shows a flashing full-screen alert, keeps a detection history and lets the user disable classes and set a global sensitivity $s \in [-1, 1]$ that scales every threshold as $\theta'_c = \mathrm{clip}(\theta_c(1-0.4s),\, 0.05,\, 0.95)$.

Privacy. Audio exists only as floating-point samples in the ring buffer and is overwritten every 0.5 s. Nothing is written to storage, the release manifest has no `INTERNET` permission, and neither audio, embeddings nor scores cross the Bluetooth link. Unlike "record and upload" designs, the system is continuously aware, yet it retains nothing.

4 Safety-Asymmetric Detection

We partition the classes into a safety-critical set $\mathcal{S} = \{$smoke alarm, siren, glass break$\}$ and the remaining notification classes $\mathcal{N}$. Asymmetry enters at four points.

4.1 Cost-Weighted Training

The head is trained with a positive-weighted binary cross-entropy

$$\mathcal{L} = -\frac{1}{C}\sum_{c=1}^{C}\Big[w_c\, y_c \log \hat{y}_c + (1-y_c)\log(1-\hat{y}_c)\Big]\,, \tag{1}$$

with $w_c = 3.0$ for $c \in \mathcal{S}$, 1.5 for baby cry and appliance beep, and 1.0 otherwise. A missed positive therefore costs up to three times as much as a false positive, following the standard cost-sensitive formulation [2].

4.2 Precision-Floored Threshold Calibration

A single 0.5 threshold is arbitrary for sigmoid outputs trained on imbalanced data. On the validation split we search $\theta \in \{0.10, 0.15, \ldots, 0.90\}$ for each class:

$$\theta_c^{*} = \begin{cases} \arg\max_{\theta} R_c(\theta) \;\text{ subject to }\; P_c(\theta) \geq \pi\,, & c \in \mathcal{S}\,, \\ \arg\max_{\theta} F_{1,c}(\theta)\,, & c \in \mathcal{N}\,, \end{cases} \tag{2}$$

with precision floor $\pi = 0.70$. For a safety class, the system should catch as many true events as possible, but not so many false ones that users learn to ignore it, which is the trust failure reported in [11]. If no threshold meets the floor, the most precise one is taken and the class is flagged as under-calibrated instead of being reported as fitted.

4.3 Asymmetric Temporal Decision Rule

Per-window scores are noisy. A class c fires at step t when its current window is a hit ($s_{c,t} \geq \theta_c$), at least k_c of its last $n = 3$ windows are hits, and it has not fired within a 10 s cooldown. We use $k_c = 1$ for $c \in \mathcal{S}$ and $k_c = 2$ otherwise.

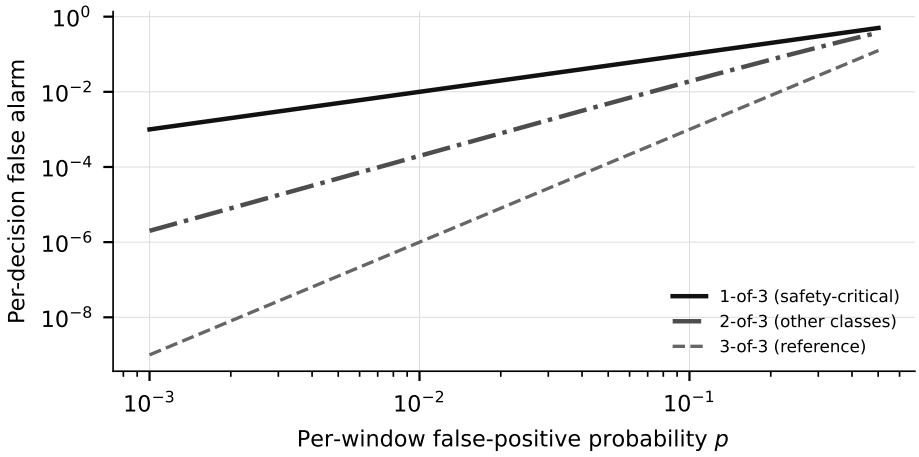


Fig. 2. Analytical per-step false-alarm probability of the decision rule against the per-window false-positive rate p, assuming independent windows. The 2-of-3 rule used for notification classes (*dash-dotted*) lowers false alarms by a factor of about $2p$ relative to the 1-of-3 rule reserved for safety-critical classes (*solid*)

False Alarms. If each background window is a false hit independently with probability p, the per-step false-alarm probability is

$$F_k(p) = p \sum_{j=k-1}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j}\,, \tag{3}$$

so $F_1 = p$ and $F_2 = p\,(1-(1-p)^2) \approx 2p^2$. At $p = 0.05$ the 2-of-3 rule reduces false alarms from 0.05 to 0.0049, a tenfold reduction, and the cooldown caps alerts at six per minute per class (Fig. 2). Real windows overlap by 50 %, so their errors are positively correlated and the true reduction is smaller; Table 6 measures it.

Latency. Let $W = 1\,\mathrm{s}$ be the window, $H = 0.5\,\mathrm{s}$ the hop and $m = 2$ the number of pooled embeddings, and assume that windows lying entirely inside a sustained event score above threshold. The first such window ends at most $W + H$ after onset, the pooled vector is free of pre-onset audio after $m - 1$ further hops, and the rule needs $k_c - 1$ further hits. The worst-case decision latency is therefore

$$L_c \le W + mH + (k_c - 1)H + \tau\,, \tag{4}$$

where τ is the per-window inference time: $2.0\,\mathrm{s} + \tau$ for safety-critical classes and $2.5\,\mathrm{s} + \tau$ otherwise. Smoothing thus adds no delay to safety classes. In practice, partial windows often clear the threshold earlier.

Table 1. Wristband bill of materials (approximate Indian retail prices, 2026)

Component	Price (Rs.)
ESP32 DevKit V1 (WROOM-32)	350
3.7 V LiPo cell, 500 mAh	180
TP4056 USB-C charger with protection	35
3.3 V low-dropout regulator	20
10 mm coin vibration motor	40
AO3400 MOSFET, diodes, resistors, LED, switch	31
Perfboard, wiring, headers	60
Strap and enclosure	150
Total	≈870

4.4 Class-Coded Haptics

Each alert reaches the wristband as four bytes: protocol version, an alert command, the class identifier and an intensity. Safety-critical classes always use full intensity (255); for the others the intensity is the confidence scaled to $[128, 255]$. The firmware maps each class to a distinct rhythm, so the user can identify the event without looking: smoke alarm uses the "temporal-three" pattern of three long pulses that standard fire alarms use [9], a siren is a rising and falling swell, a doorbell is a strong then soft "ding-dong", and knocking is four quick taps. A new alert interrupts the current one, and dismissing it on the phone sends a stop command.

5 Wristband Hardware

The wristband is an ESP32-WROOM-32 Bluetooth Low Energy peripheral with a 10 mm coin vibration motor driven through a logic-level MOSFET with 20 kHz PWM, a 500 mAh LiPo cell, a protected TP4056 charger and a low-dropout 3.3 V regulator. Its bill of materials totals about Rs. 870 at 2026 Indian retail prices (Table 1). To save power, the firmware clocks the CPU at 80 MHz, transmits at +3 dBm and requests a 30–50 ms connection interval with slave latency 2, which targets an alert delivery delay below about 150 ms. When disconnected it advertises every 100 ms for 30 s and every second afterwards. Below 20 % charge it lowers motor strength and transmit power, and at 3.4 V it enters deep sleep. When the phone disconnects, two long buzzes tell the user that alerts have stopped, so a silent wristband never looks the same as a quiet room.

6 Data and Evaluation Protocol

6.1 Public Data

Target clips come from FSD50K [4] (4,494 clips mapped from its ground truth) and AudioSet [5] (161 downloaded clips, 20–21 per class); background clips come

Table 2. Clips per class and source. Public counts come from the committed manifests and SAHARA-IN counts from the field-recording metadata; † marks the safety-critical classes $\mathcal{S}$

Class	FSD50K	AudioSet	SAHARA-IN	Total
Smoke alarm†	0	20	0	20
Doorbell	144	20	16	180
Siren†	132	20	0	152
Knocking	373	29	14	416
Dog bark	536	20	11	567
Baby cry	0	20	6	26
Glass break†	1,241	34	0	1,275
Appliance beep	2,068	25	16	2,109
Background	DESED 905	–	60	965

from DESED real soundscapes [22] (905 clips). FSD50K has no smoke-alarm or baby-cry label, so these two classes have only 20 AudioSet clips each, and the label distribution is highly skewed (2,093 appliance-beep clips against 20 smoke-alarm clips; Table 2). Splits are 70/15/15, stratified by class and grouped by recording identifier so that segments of one recording never cross splits.

6.2 Synthetic Indian-Noise Mixtures

To model Indian acoustic conditions at scale, each training target clip is mixed with three random 5 s chunks of iNoise [15], which covers autorickshaws, buses, railway stations, streets, homes and workplaces, or of DESED, at an SNR drawn uniformly from $[-5, 10]$ dB. Mixtures are tagged `synthetic_mixed` in every manifest so that real and synthetic results are never conflated.

6.3 SAHARA-IN Field Recordings

Four team members recorded 123 clips (14.6 min) on their own smartphones: 63 target-class clips (16 appliance beep, 16 doorbell, 14 knocking, 11 dog bark and 6 baby cry) and 60 background clips. The backgrounds include street hawkers, a temple, a domestic kitchen, television, monsoon rain and hard negatives such as drawers, doors and a washing-machine drum spinning without beeping. Across all 123 clips, the recording locations are apartments (81), kitchens (11), temples (11), street markets (10) and outdoor rain (10). We split by recorder: clips from two recorders (66) are used for training, and those from the other two (57: 34 target, 23 background) are held out, so the test set measures generalisation to unseen people, phones and homes. A separate set of 57 name-call clips (38 positive, 19 negative) supports the planned personalisation module (Sect. 8).

6.4 Metrics

We report per-class average precision (AP) and its macro mean (mAP), as well as precision, recall and F_1 at both the default 0.5 and the calibrated thresholds.

Table 3. Baseline test results on 698 held-out clips at a 0.5 threshold

Class	Precision	Recall	F_1	AP
Smoke alarm†	0.000	0.000	0.000	0.090
Doorbell	0.682	0.395	0.500	0.632
Siren†	0.333	0.333	0.333	0.250
Knocking	0.826	0.655	0.731	0.751
Dog bark	1.000	0.893	0.943	0.956
Baby cry	0.000	0.000	0.000	0.669
Glass break†	0.881	0.895	0.888	0.937
Appliance beep	0.860	0.790	0.824	0.923
Macro average	0.573	0.495	0.527	0.651

For streaming behaviour we report false alarms per hour (FA/h) on continuous background audio (DESED, iNoise and SAHARA-IN backgrounds), and we measure latency and energy on a phone. Model selection and calibration use only the validation split.

7 Results

7.1 Baseline on Public Data

Table 3 and Fig. 3 report the first full baseline, trained on 3,259 clip-level embeddings and tested on 698 held-out FSD50K and AudioSet clips, without augmentation. Early stopping ended training after 9 epochs, with a final validation accuracy of 0.810. The model reaches 0.651 mAP. Classes with hundreds of FSD50K clips perform well (dog bark 0.956, glass break 0.937 and appliance beep 0.923 AP). The two AudioSet-only classes fail at the default threshold: smoke alarm reaches only 0.090 AP, and baby cry has zero recall at 0.5 despite 0.669 AP. Each has three test clips, so their per-class estimates carry wide uncertainty. Siren, a safety-critical class with 152 clips, reaches only 0.250 AP. Two lessons follow. For safety-critical classes, the bottleneck is data rather than model capacity. And mAP alone hides the failures that matter most to the user, which motivates per-class, safety-weighted reporting.

7.2 Effect of Threshold Calibration

Table 4 shows the calibrated thresholds on the validation split. Calibration lowers most thresholds well below 0.5 and raises recall: knocking from 0.390 to 0.576, doorbell from 0.304 to 0.500, and baby cry from 0.667 to 1.000 at unchanged precision. Glass break trades precision (0.830 to 0.714) for recall (0.852 to 0.926) while keeping precision above the 0.70 floor, which is exactly what the constraint is designed to do. For smoke alarm, no threshold met the floor (best precision 0.25), so the calibrator flags the class as under-calibrated instead of hiding the problem. These are validation-set figures; the corresponding test-set results are in Table 5.

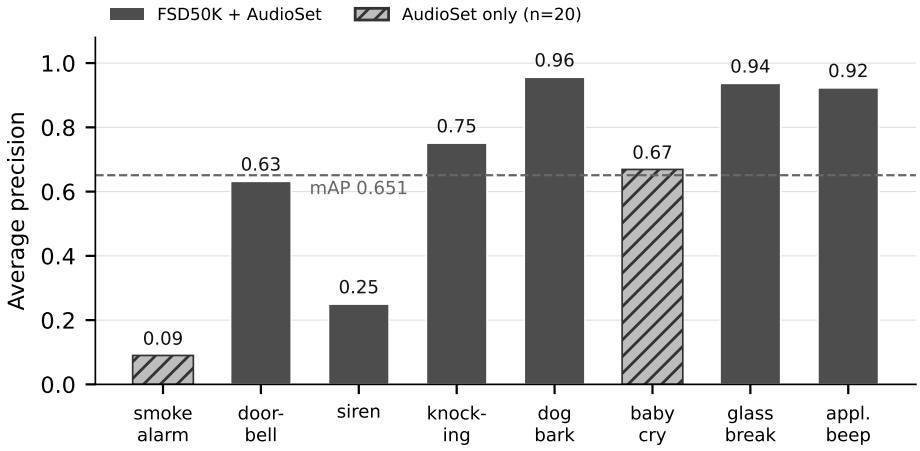


Fig. 3. Per-class test AP of the baseline. The *hatched* classes have only 20 public clips each (AudioSet) and three test clips each; the *dashed line* marks the mAP

Table 4. Per-class threshold calibration on the validation split: precision (P) and recall (R) at the default 0.5 and at the calibrated threshold θ^*

Class	θ^*	P at 0.5	P at θ^*	R at 0.5	R at θ^*
Smoke alarm†	0.15	0.000	0.250	0.000	0.333
Doorbell	0.20	0.583	0.426	0.304	0.500
Siren†	0.40	1.000	1.000	0.333	0.333
Knocking	0.15	0.719	0.680	0.390	0.576
Dog bark	0.30	0.972	0.960	0.854	0.878
Baby cry	0.10	1.000	1.000	0.667	1.000
Glass break†	0.25	0.830	0.714	0.852	0.926
Appliance beep	0.50	0.858	0.858	0.847	0.847

7.3 Full Model and Generalisation to Indian Conditions

Table 5 reports the final model, trained at window level (1 s windows, matching on-device inference) on public data, synthetic Indian-noise mixtures and the SAHARA-IN training recorders. It is evaluated on test sets of increasing difficulty: the public held-out split, synthetic mixtures at fixed SNRs and the recorder-disjoint SAHARA-IN split.

[TBD: 2–3 sentences interpreting Table 5: the drop from public data to SAHARA-IN, which classes transfer, and FA/h against a target of fewer than one alert per hour.]

Table 5. Final model at calibrated thresholds: mAP, macro recall, mean recall over the safety-critical classes and false alarms per hour (FA/h)

Test set	mAP	Macro R	Safety R	FA/h
Public held-out split	**TBD**	**TBD**	**TBD**	–
Synthetic mixture, SNR 10 dB	**TBD**	**TBD**	**TBD**	–
Synthetic mixture, SNR 0 dB	**TBD**	**TBD**	**TBD**	–
Synthetic mixture, SNR −5 dB	**TBD**	**TBD**	**TBD**	–
SAHARA-IN, unseen recorders	**TBD**	**TBD**	–	**TBD**
Background stream (DESED, iNoise)	–	–	–	**TBD**

Table 6. Ablation on the public test split, with FA/h measured on continuous background audio

Configuration	mAP	Safety R	FA/h
Unweighted loss, $\theta = 0.5$, no smoothing	**TBD**	**TBD**	**TBD**
+ cost-weighted loss	**TBD**	**TBD**	**TBD**
+ precision-floored calibration	**TBD**	**TBD**	**TBD**
+ 2-of-3 rule for all classes	**TBD**	**TBD**	**TBD**
+ asymmetric 1-of-3 / 2-of-3 rule (SAHARA)	**TBD**	**TBD**	**TBD**

7.4 Ablation of the Asymmetric Components

Table 6 adds the safety-asymmetric components one at a time, measured on the public test split (safety-class recall) and on continuous background audio (FA/h).

7.5 On-Device Cost

Table 7 reports the deployment cost on **[TBD: phone model and Android version]**. The classifier head adds 296,328 parameters to the roughly 3.7 million of YAMNet [20].

8 Discussion and Limitations

Data Scarcity for the Most Critical Classes. The baseline shows that smoke alarm and baby cry, which carry the highest stakes, have the least public data. SAHARA-IN does not yet contain smoke-alarm, siren or glass-break recordings, because they are hard to record safely. Targeted collection, for example a controlled smoke-alarm test with the building's consent, is our most valuable next step.

Small Field Set. With 123 clips and a recorder split that leaves some classes on only one side, SAHARA-IN measures transfer for doorbell, dog bark and appliance beep, plus false alarms on Indian backgrounds, but it cannot yet estimate every class. We release the protocol so that the set can grow.

Table 7. On-device cost

Measure	Value
Head parameters	296,328
TFLite size, FP32 / INT8 (YAMNet and head) (MB)	**TBD** / **TBD**
INT8 change in mAP relative to FP32	**TBD**
Inference time per window, median / 95th percentile (ms)	**TBD** / **TBD**
End-to-end latency, sound onset to phone alert (s)	**TBD**
Phone alert to wristband vibration (ms)	**TBD**
Phone battery drain with the service running (%/h)	**TBD**
Wristband current, connected and idle (mA)	**TBD**
Wristband battery life, 500 mAh (h)	**TBD**

Independence Assumption. The false-alarm analysis assumes independent windows. Overlapping windows are correlated, so the analysis is an upper bound on the benefit; Table 6 reports the measured effect.

No User Study Yet. We have not yet evaluated SAHARA with DHH participants. Prior work shows that perceived usefulness depends on false-alarm rates and on notification design [3, 11]. A field deployment with DHH users in Mumbai, measuring missed events, alert fatigue and recognition of the haptic patterns, is planned.

Personalisation. Users want to be alerted when someone calls their name. We have collected a name-call set and plan to fuse a few-shot prototypical module [13, 21] with the main head, so that users can enrol their own name and household sounds on the device.

9 Conclusion

SAHARA shows that an assistive sound-awareness system can encode the unequal cost of missed safety sounds at every stage, from the loss function to the vibration pattern, while running entirely on a phone and a Rs. 870 wristband and retaining no audio. Our baseline results show that mAP alone hides the failures that matter most, and that the classes users depend on most have the least data. We release the code, the Bluetooth protocol, the firmware and the SAHARA-IN evaluation protocol to support safety-aware, privacy-preserving and regionally representative assistive audio research.

Acknowledgements. **[TBD: thank the guide and department; mention any funding]**

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